

Groups and Lie Algebras: Exercise Sheet 5

1. 2-dimensional lattices

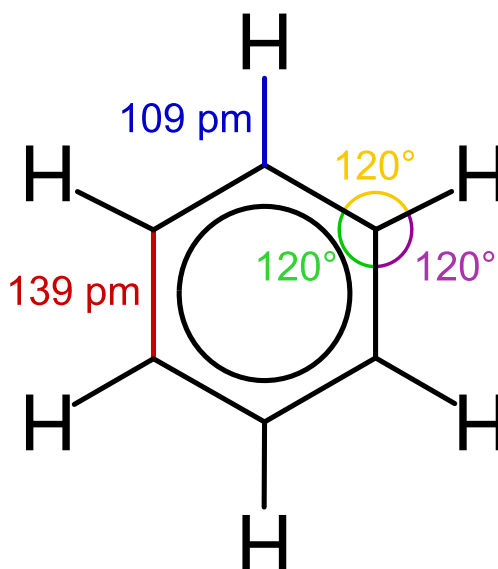
Consider the group $\mathbb{Z}^2$ encoding the translation symmetry of a lattice. Its group of automorphisms is $\text{Gl}(2, \mathbb{Z})$ of matrices with integer coefficients that have an inverse with integer coefficients. The point groups of the rhombic and rectangular lattice are both $\mathbb{Z}_2 \times \mathbb{Z}_2$.

- Compute the group homomorphism $\mathbb{Z}_2 \times \mathbb{Z}_2 \rightarrow \text{Gl}(2, \mathbb{Z})$ describing the full symmetry group as a semi-direct product in both cases.
- Show that the full symmetry group of the rhombic and rectangular lattice aren't isomorphic.

2. Molecule and selection rules

Consider the water molecule H_2O . As discussed in the lecture, its symmetry group is C_{2v} in Schönflies notation. Find selection rules for the matrix elements of the operators x, y, z .

Determine the symmetry group of the benzene molecule C_6H_6 :



The unlabelled vertices are the location of the C atoms. Picture taken from: [https://en.wikipedia.org/wiki/Benzene#/media/File:Benzene_geometrie_\(2\).svg](https://en.wikipedia.org/wiki/Benzene#/media/File:Benzene_geometrie_(2).svg)

3. 1D band structure from translation symmetry Consider a particle on a one-dimensional lattice with sites labeled by $n \in \mathbb{Z}$. The Hilbert space has an orthonormal basis $\{|n\rangle\}_{n \in \mathbb{Z}}$, and the Hamiltonian is

$$H = -t \sum_{n \in \mathbb{Z}} (|n\rangle \langle n+1| + |n+1\rangle \langle n|),$$

- (a) Define the translation operator T by

$$T |n\rangle = |n+1\rangle.$$

Show that T is unitary and that $[H, T] = 0$. Conclude that H can be diagonalized in a basis of eigenvectors of T .

- (b) The group generated by T is isomorphic to $\mathbb{Z}$. Its irreducible unitary representations are labeled by $k \in [-\pi, \pi)$ and given by

$$\chi_k(n) = e^{ikn}.$$

Define the corresponding states

$$|k\rangle = \sum_{n \in \mathbb{Z}} e^{ikn} |n\rangle.$$

Show that

$$T |k\rangle = e^{ik} |k\rangle.$$

- (c) Compute $H |k\rangle$ and show that $|k\rangle$ is an eigenvector of H with eigenvalue

$$E(k) = -2t \cos k.$$

- (d) Plot $E(k)$ for $k \in [-\pi, \pi]$.